Physics of Fluids

RESEARCH ARTICLE | AUGUST 01 1994

Negative viscosity of ferrofluid under alternating magnetic field

Mark I. Shliomis; Konstantin I. Morozov

Check for updates

Physics of Fluids 6, 2855–2861 (1994)
https://doi.org/10.1063/1.868108

View Online
Export Citation

Articles You May Be Interested In

Wavelength dependence of the Faraday effect and magnetobirefringence in ferrofluid thin films
J. Appl. Phys. (May 1993)

A new look at the Bitter method of magnetic imaging
Rev. Sci. Instrum. (March 1991)

Ferrofluid studies of recorded data and defect identification in small high-performance rigid discs
J. Appl. Phys. (April 1985)

16 March 2026 08:59:36

AIP Advances
Why Publish With Us?
21DAYS average time to 1st decision
OVER 4 MILLION views in the last year
INCLUSIVE scope
Learn More
AIP Publishing

AIP Publishing

Negative viscosity of ferrofluid under alternating magnetic field

Mark I. Shliomis[a] and Konstantin I. Morozov
Institute of Continuous Media Mechanics of the Russian Academy of Sciences, Perm' 614061, Russia

(Received 17 September 1993; accepted 19 April 1994)

A stationary magnetic field induces an increase in the ferrofluid viscosity. An additional resistance to the flow occurs due to the field oriented magnetic particles impeded by free rotation in a vortex flow. It is shown that in an alternating, linearly polarized magnetic field the additional viscosity is positive at low frequencies of the field and negative at high frequencies. The point is that an alternating field induces rotatory oscillations of the particles, but does not single out any direction of their rotation. One can say that half of the particles rotate clockwise and the other half counterclockwise. Hence, the macroscopic angular velocity of the particles equals zero. However, this corresponds only to fluid at rest. Any shear (i.e., any vorticity) is sufficient to break down the degeneracy of the direction of rotation, which results in the nonzero angular velocity of the particles. The occurring "spin up" of the flow by the rotating particles leads to the decrease of the effective viscosity, which means the additional viscosity appears to be negative.

I. INTRODUCTION

It is known[1] that in the presence of a stationary magnetic field the ferrofluid viscosity is increased. Let us recall the mechanism of this effect.[2] Moving with a velocity $\mathbf{v}$, each element of the liquid volume rotates with a local angular velocity $\mathbf{\Omega}=\nabla\times\mathbf{v}/2$. In the absence of the magnetic field the colloidal particles of the ferromagnet included in this volume rotate with the same velocity: $\mathbf{\Theta}=\mathbf{\Omega}$, where $\mathbf{\Theta}$ is the macroscopic angular velocity of the rotating particles, i.e., averaged over the physically small elements of volume. Any deviation of $\mathbf{\Theta}$ from $\mathbf{\Omega}$ decays to zero in a very short time

$$\tau_s=\rho_s d^2/60\,\eta. \tag{1}$$

Assuming the diameter of the particle to be $d=10^{-6}$ cm, the density of a particle material $\rho_s=6$ g/cm^3, and the viscosity of the liquid $\eta=10^{-2}$ g/cm s, we obtain $\tau_s=10^{-11}$ s.

The situation is changed in the presence of a stationary magnetic field. The latter causes a partial orientation of magnetic moments of the particles, thereby preventing their free rotation in vortical flow. Maintaining this difference of angular velocities $\mathbf{\Theta}-\mathbf{\Omega}$, the field induces an additional dissipation of the kinetic energy of the fluid. For the additional (rotational or vortical) viscosity a simple expression was derived by Shliomis:[2]

$$\Delta\eta=\frac{3}{2}\,\eta\varphi\frac{\xi-\tanh\xi}{\xi+\tanh\xi}\sin^2\beta, \tag{2}$$

where $\xi=mH/kT$ is the Langevin parameter (the ratio of an energy of the particle's magnetic moment $\mathbf{m}$ in the field $\mathbf{H}$ to the thermal energy), $\varphi=nV$ is the particle volume concentration, and β is the angle between the vectors $\mathbf{H}$ and $\mathbf{\Omega}$. As can be seen from (2), if $\mathbf{H}$ is parallel to $\mathbf{\Omega}$, the viscosity is independent of the field. The orientation of the particle magnetic moments along the field allows the particles to rotate freely with the same angular velocity $\mathbf{\Omega}$ of the fluid. Below we will assume that $\beta=\pi/2$; this situation (with $\mathbf{H}\perp\mathbf{\Omega}$) occurs for example in a plane Couette or Poiseuille flow, when the field is oriented perpendicularly to the flow (i.e., to the planes of the shear), or in a cylindrical Poiseuille flow when capillary tube is aligned along the axis of the magnetic coil.

According to (2), in the low stationary field, $\Delta\eta$ grows proportionally to ξ^2:

$$\Delta\eta=\tfrac{1}{4}\eta\varphi\xi^2,\quad \xi\ll 1 \tag{3}$$

and tends to the saturation value in the strong field:

$$\Delta\eta=\tfrac{3}{2}\eta\varphi,\quad \xi\gg 1. \tag{4}$$

It is interesting to note that both the rotational viscosity and the supplement $5\eta\varphi/2$ in the Einstein' formula for "translational" viscosity of suspension have the same order of magnitude. So it is clear that in our case we are dealing with viscosity of diluted suspensions, when one can neglect both the hydrodynamic and magnetic interparticle interactions. Note that numerous experimental data for the ferrofluid viscosity in a stationary magnetic field (first measurements have been made almost a quarter of a century ago[3]) are in good agreement with Eq. (2).

It is curious that for all these years nobody has been interested in the viscosity of ferrofluid in an alternating, linearly polarized magnetic field $\mathbf{H}=(H_0\cos\omega t,0,0)$. One can guess, however, why that happened. It should be remarked that, when field is shifted in direction the particle magnetization attempts to regain equilibrium and can do so by one of the two processes.[4] In Brownian relaxation the particle rotates together with its magnetic moment fixed relative to the crystal axis of the particle. The rotation is resisted by viscous torque due to surrounding carrier liquid and characterized by Brownian time of rotational diffusion

$$\tau_B=3\,\eta V/kT, \tag{5}$$

where $V=\pi d^3/6$ is the volume of a particle. In the Neel mechanism the magnetic moment rotates relative to the crystal axis so that the particle itself does not rotate. It is for this reason that only the particles with the Brownian mechanism of magnetization relaxation (i.e., with "freezing in" mag-

[a]Present address: Department of Physics of Complex Systems, Weizmann Institute of Science, Rehovot 76100, Israel.

 1070-6631/94/6(8)/2855/7/$6.00 © 1994 American Institute of Physics

16 March 2026 08:59:36

netic moments) contribute to the magnetic viscosity of ferrofluids. Recall that the condition of freezing takes place only for big enough particles while the Neel relaxation (it is very sensitive to the particle size) dominates in small particles. The crossover particle size d_s at which both effects are comparable in rate[4] occurs near the size of particles commonly making up ferrofluids (this characteristic diameter depends on the sort of ferromagnet and the fluid viscosity). At $d>d_s$ magnetization relaxation occurs during the Brownian time (5). But the existence of a finite relaxation time means that the orienting influence of alternating field decreases with increasing frequency and tends to zero for $\omega\tau_B \gg 1$. In the latter case, ferrocolloid does not have enough time for remagnetization. The reduction of magnetization has to be accompanied by reduction of rotational viscosity as well. Generally, at the limit $\omega\tau_B \to \infty$ the particles cease to feel the magnetic field. Therefore, it seems that the case of an alternating field is of no interest for the calculation of ferrofluid viscosity. According to the statement above it was expected that the dependence of $\Delta\eta(\omega)$ will turn out a trivial one: starting from the value (2) at zero frequency, $\Delta\eta$ will monotonically diminish to zero as ω tends to infinity.

Solution of the ferrohydrodynamic equations shows, however, that in reality the viscosity has a much more interesting behavior. According to our calculations, the vortical viscosity quickly falls with the growth of frequency ω and at a certain value ω_0 (it depends on the amplitude of magnetic field) $\Delta\eta$ changes its sign passing from region of positive values to region of negative ones. In the latter the function $\Delta\eta(\omega)$ attains its minimum and after that begins to grow up. As $\omega\tau_B$ increases the viscosity remains negative and at the limit $\omega\tau_B \to \infty$ the rotational viscosity indeed tends to zero.

The above unexpected result can be obtained analytically in the case of a low field, i.e., at $\xi \ll 1$. The solution of the problem and discussion of a physical mechanism of negative viscosity are given in Sec. II. The results of solution for arbitrary values of the field amplitude are presented in Sec. III. In the concluding section we discuss the possibility to obtain the negative total viscosity: $\eta+\Delta\eta<0$.

II. ROTATIONAL VISCOSITY UNDER THE LOW-AMPLITUDE ALTERNATING FIELD

The system of ferrohydrodynamic equations accounting for the connection between the magnetic and rotational mechanical degrees of freedom consists of the fluid motion equation, the magnetization equation, and the equation of rotational motion of the particles:[2]

$$\rho\frac{d\mathbf{v}}{dt} = -\nabla p + (\mathbf{M}\cdot\nabla)\mathbf{H} + \eta\Delta v + \frac{I}{2\tau_s}\nabla\times(\boldsymbol{\Theta}-\boldsymbol{\Omega}), \quad (6)$$

$$\frac{d\mathbf{M}}{dt} = \boldsymbol{\Theta}\times\mathbf{M} - \frac{1}{\tau_B}(\mathbf{M}-\mathbf{M}_0), \quad (7)$$

$$I\frac{d\boldsymbol{\Theta}}{dt} = \mathbf{M}\times\mathbf{H} - \frac{I}{\tau_s}(\boldsymbol{\Theta}-\boldsymbol{\Omega}). \quad (8)$$

The complete set of equations includes also the equations

$$\nabla\cdot\mathbf{v}=0, \quad \nabla\times\mathbf{H}=0, \quad \nabla\cdot(\mathbf{H}+4\pi\mathbf{M})=0, \quad (9)$$

indicating that the fluid is considered to be incompressible and nonconducting.

In the above equations I is the sum of particles' moment of inertia over the unit volume, $\mathbf{M}_0$ is the "instantaneous equilibrium" magnetization which would exist in a given $\mathbf{H}(t)$ at $\tau_B=0$, i.e., if the magnetization followed the field without retardation. Evidently, the dependency $\mathbf{M}_0[\mathbf{H}(t)]$ is determined by the same Langevin formula as in the case of the constant field:

$$\mathbf{M}_0 = nmL(\xi)\mathbf{H}/H,$$
$$\xi = mH(t)/kT, \quad (10)$$
$$L(\xi) = \coth\xi - \xi^{-1}.$$

Indeed, if the magnetization relaxation time equals to zero, the magnetic field will be a "slowly varying function" of time for any dependency $\mathbf{H}(t)$. In order to avoid a misunderstanding we note that the real magnetization $\mathbf{M}(t)$ differs from $\mathbf{M}_0(t)$ simply because of the nonzero relaxation time τ_B in Eq. (7).

Since τ_s is small [see Eq. (1)] the inertial term in Eq. (8) is negligible in comparison with the relaxation term: $I\,d\boldsymbol{\Theta}/dt \ll I\boldsymbol{\Theta}/\tau_s$, so Eq. (8) has the form

$$\boldsymbol{\Theta} = \boldsymbol{\Omega} + (\tau_s/I)(\mathbf{M}\times\mathbf{H}). \quad (11)$$

Now one can eliminate the difference $\boldsymbol{\Theta}-\boldsymbol{\Omega}$ from Eqs. (6) to (7):

$$\rho\frac{d\mathbf{v}}{dt} = -\nabla p + (\mathbf{M}\cdot\nabla)\mathbf{H} + \eta\Delta v + \frac{1}{2}\nabla\times(\mathbf{M}\times\mathbf{H}), \quad (12)$$

$$\frac{d\mathbf{M}}{dt} = \boldsymbol{\Omega}\times\mathbf{M} - \frac{1}{\tau_B}(\mathbf{M}-\mathbf{M}_0) - \frac{\tau_s}{I}\mathbf{M}\times(\mathbf{M}\times\mathbf{H}). \quad (13)$$

The system of the ferrohydrodynamic equations (6)–(10) or the equivalent system (10)–(13) allows the calculation of the effective viscosity of the ferrofluid in the stationary[2] and alternating magnetic field.

Thus the problem is to express the viscosity $\Delta\eta$ as a function of amplitude and frequency of the field

$$H_x = H_0\cos\omega t, \quad H_y = H_z = 0. \quad (14)$$

In this section we will assume that the field amplitude is small in the sense $mH_0 \ll kT$. This assumption permits to evade the difficulties which are connected with the nonlinearity of the Langevin function (10), instead of which we now have

$$\mathbf{M}_0(t) = \chi\mathbf{H}(t), \quad \chi = nm^2/3kT. \quad (15)$$

We can therefore obtain an analytical solution of the problem. Below we give two variants of its solution. In the first the linear-polarized field (14) is presented as a sum of two fields, which are circularly polarized opposite each other. This approach is rather visual and in our opinion it makes clear the physical nature of the problem. The second version of the solution is simpler but more formal.

16 March 2026 08:59:36

A. First variant

Consider the magnetic field (14) as a superposition of two rotating fields—the left-hand polarized field (subscript +) and the right-hand polarized one (−):

$$\mathbf{H}_{\pm}=(H_0\cos\omega t,\pm H_0\sin\omega t,0),\quad \mathbf{H}=\tfrac{1}{2}(\mathbf{H}_{+}+\mathbf{H}_{-}). \tag{16}$$

For each of these fields taken separately, Eqs. (6)–(8) admit a solution, in which the fluid is quiescent ($\mathbf{\Omega}=0$) and the magnetization rotates with an angular frequency of the field, lagging behind it at some angle α. The ferroparticles rotate with a stationary angular velocity, which is oriented perpendicularly to the plane of the field rotation:

$$\begin{aligned}&\mathbf{M}_{\pm}=[M\cos(\omega t-\alpha),\pm M\sin(\omega t-\alpha),0],\\&\mathbf{\Theta}_{\pm}=(0,0,\pm\Theta).\end{aligned} \tag{17}$$

Substituting (15)–(17) into Eqs. (7) and (8), we obtain

$$\begin{aligned}&M=\chi H_0\cos\alpha,\\&\Theta=(\tau_s/I)MH_0\sin\alpha,\\&\tan\alpha=(\omega-\Theta)\tau_B.\end{aligned} \tag{18}$$

Since $\xi=mH_0/kT\sqrt{2}$ is small (we defined here the Langevin parameter over the "acting" magnetic field $H_0/\sqrt{2}$), using the expression $I/\tau_s=6\eta\varphi$, one can eliminate the angle α from the solution (18). Up to the ξ^2 terms we have

$$\begin{aligned}&M=\frac{\xi H_0}{\sqrt{1+\omega^2\tau_B^2}},\\&\Theta=\omega\frac{\xi^2/3}{1+\omega^2\tau_B^2},\\&\tan\alpha=\omega\tau_B\left(1-\frac{\xi^2/3}{1+\omega^2\tau_B^2}\right).\end{aligned} \tag{19}$$

As one can see from (17), in the field $\mathbf{H}=(\mathbf{H}_{+}+\mathbf{H}_{-})/2$ directed along the axis x the velocity of the particles rotation Θ_z vanishes together with the component of magnetization M_y perpendicular to the field. The reason is clear: the alternating field with linear polarization does not single out any preferable direction of the particles rotation. One can say that at each single moment of time one half of the particles make rotational swings clockwise and the other half rotate counterclockwise. Therefore, the macroscopic (i.e., average) velocity equals zero.

The degeneracy of the direction of rotation takes place only in the case of motionless fluid. Any flow with vorticity $\nabla\times\mathbf{v}=2\mathbf{\Omega}$, caused for example by the motion of the boundaries in case of the Couette flow or by the pressure gradient in the Poiseuille flow, eliminates this degeneracy because it destroys the symmetry between the "plus" and "minus" solutions. Let us show this.

Let us study the variation of the solution (17) in the presence of the hydrodynamical vortex $\mathbf{\Omega}=(0,0,\Omega)$. According to (11), the rotation rates in the left- and right-polarized fields prove to be different, so that instead of Eq. (18) we obtain

$$\begin{aligned}&M_{+}=\chi H_0\cos\alpha_{+},\\&\Theta_{+}=(\tau_s/I)M_{+}H_0\sin\alpha_{+}+\Omega,\\&\tan\alpha_{+}=(\omega-\Theta_{+})\tau_B;\\&M_{-}=\chi H_0\cos\alpha_{-},\\&\Theta_{-}=(\tau_s/I)M_{-}H_0\sin\alpha_{-}-\Omega,\\&\tan\alpha_{-}=(\omega-\Theta_{-})\tau_B.\end{aligned} \tag{20}$$

It can be seen that the rotation of the particles is faster when the field and the fluid are rotating in the same direction (that is, when both ω and Ω have the same sign). Retaining only the principal terms in the expression for $\tan\alpha_{\pm}$ in (20), we have

$$\tan\alpha_{+}=(\omega-\Omega)\tau_B,\quad \tan\alpha_{-}=(\omega+\Omega)\tau_B. \tag{21}$$

Owing to the difference between the angles α_{+} and α_{-}, the superposition of the fields $\mathbf{H}_{+}$ and $\mathbf{H}_{-}$ generates the nonzero y component of magnetization, contrary to the case of quiescent fluid:

$$\begin{aligned}M_y^{(+)}&=\chi H_0\cos\alpha_{+}\sin(\omega t-\alpha_{+})\\&=\frac{\chi H_0}{1+(\omega-\Omega)^2\tau_B^2}[\sin\omega t-(\omega-\Omega)\tau_B\cos\omega t],\\M_y^{(-)}&=-\chi H_0\cos\alpha_{-}\sin(\omega t-\alpha_{-})\\&=-\frac{\chi H_0}{1+(\omega+\Omega)^2\tau_B^2}[\sin\omega t-(\omega+\Omega)\tau_B\cos\omega t].\end{aligned} \tag{22}$$

The Brownian rotational diffusion time for the particles with diameter 10^{-6} cm is usually not higher than 10^{-5} s. Therefore the inequality $\Omega\tau_B\ll 1$ is almost always valid (it may be violated only in highly viscous fluids). If the field is linearly polarized along the x direction, the y component of magnetization (in the linear with respect to $\Omega\tau_B$ approximation) is

$$M_y=\tfrac{1}{2}(M_y^{(+)}+M_y^{(-)})=\Omega\tau_B\chi H_0\cos^2\alpha\cos(\omega t-2\alpha). \tag{23}$$

Here α is the phase difference between the magnetization and the field in the fluid at rest. It is determined from the expression

$$\tan\alpha=\omega\tau_B, \tag{24}$$

which is obtained from (19) after discarding the ξ^2 term. The last procedure is necessary because in the presence of nonlinear terms the half of the sum $\mathbf{M}_{+}+\mathbf{M}_{-}$ is not equal to the magnetization in the linearly polarized field. The principle of superposition is valid only within the framework of the linear response.

The presence of the component of magnetization (23), perpendicular to the field means that a magnetic torque acts upon the fluid. It is directed along the z axis and has the density

16 March 2026 08:59:36

$$\mathbf{M}\times\mathbf{H}=-\boldsymbol{\Omega}\tau_B\chi H_0^2\cos^2\alpha\cos\omega t\cos(\omega t-2\alpha)$$
$$=-2\boldsymbol{\Omega}\eta\varphi\xi^2\cos^2\alpha(\cos^2\omega t\cos 2\alpha+\sin\omega t\cos\omega t\sin 2\alpha). \quad (25)$$

It should be noted that the expression (25) contains two time scales: the "fast" one and the "slow" one. The fast time enters the expression explicitly, as a product with ω. The characteristic scale for ω is τ_B^{-1} as it can be seen from Eqs. (7) and (24). One can say of the low-frequency magnetic field at $\omega\tau_B\ll 1$ and the high-frequency field at $\omega\tau_B\gg 1$. The slow time is contained implicitly in the vorticity $\boldsymbol{\Omega}$, determined by hydrodynamics. But the characteristic hydrodynamic time scale $\tau_h\sim l^2/\nu$ (here l is the linear length scale of the flow, and $\nu=\eta/\rho$ is the kinematic viscosity) is always much larger than the time of the magnetization relaxation: $\tau_h\gg\tau_B$. Thus one may present the solution $\mathbf{v}$ and p (recall that $\boldsymbol{\Omega}=\nabla\times\mathbf{v}/2$) of Eqs. (12) and (13) in the form $\mathbf{v}_0+\mathbf{v}_1$ and p_0+p_1, respectively, where $\mathbf{v}_0$ and p_0 are the functions which undergo slow changes while $\mathbf{v}_1$ and p_1 are the fast-oscillating (with frequency ω) small additions. Such resolution into slow and fast changing components are not required, however, for $\mathbf{M}$ and $\mathbf{H}$ because they are the fast-oscillating functions of time without any slow-changing parts. With the first-order accuracy by $\mathbf{v}_1$ and p_1 we get from (12)

$$\rho\left(\frac{d\mathbf{v}_0}{dt}+\frac{d\mathbf{v}_1}{dt}+(\mathbf{v}_1\cdot\nabla)\mathbf{v}_0\right)$$
$$=-\nabla(p_0+p_1)+(\mathbf{M}\cdot\nabla)\mathbf{H}+\eta\Delta(\mathbf{v}_0+\mathbf{v}_1)+\frac{1}{2}\nabla\times(\mathbf{M}\times\mathbf{H}),$$

where $d/dt=\partial/\partial t+(\mathbf{v}_0\cdot\nabla)$. Averaging this equation over the fast time, i.e., over the period of the field variation $2\pi/\omega$, we have to consider $\mathbf{v}_0$ and p_0 as certain constants. As a result of the averaging, the linear by $\mathbf{v}_1$ and p_1 terms vanish because their period is equal to the field one. Omitting the subscript zero at $\mathbf{v}_0$ and p_0 we obtain

$$\rho\frac{d\mathbf{v}}{dt}=-\nabla p+\overline{(\mathbf{M}\cdot\nabla\mathbf{H})}+\eta\Delta\mathbf{v}+\frac{1}{2}\nabla\times\overline{(\mathbf{M}\times\mathbf{H})}. \quad (26)$$

While averaging (25) the first bracketed term yields $(1/2)\cos 2\alpha$, and the second term vanishes. Thus we find

$$\overline{\mathbf{M}\times\mathbf{H}}=-\boldsymbol{\Omega}\eta\varphi\xi^2\cos^2\alpha\cos 2\alpha.$$

Substituting this into (26) we see that the last term may be grouped with the viscous one:

$$\eta\Delta\mathbf{v}-\tfrac{1}{2}\eta\varphi\xi^2\cos^2\alpha\cos 2\alpha\nabla\times\boldsymbol{\Omega}$$
$$=-2(\eta+\tfrac{1}{4}\eta\varphi\xi^2\cos^2\alpha\cos 2\alpha)\nabla\times\boldsymbol{\Omega}.$$

The value, which is added to viscosity, should be considered as the additional (rotational) viscosity

$$\Delta\eta=\tfrac{1}{4}\eta\varphi\xi^2\cos^2\alpha\cos 2\alpha. \quad (27)$$

Substituting into (27) the expression for α from (24), along with $\cos 2\alpha=\cos^2\alpha-\sin^2\alpha$, we obtain

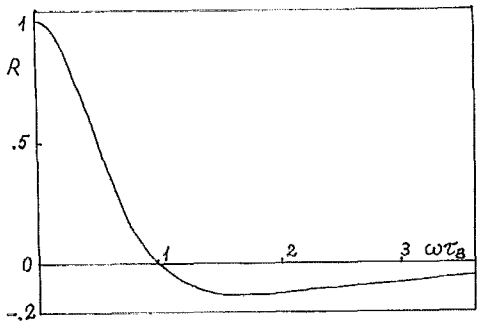


FIG. 1. The frequency dependence of the relative rotational viscosity $R=4\Delta\eta/(\eta\varphi\xi^2)$ for a low field ($\xi\ll 1$).

$$\Delta\eta=\frac{1}{4}\eta\varphi\xi^2\frac{1-\omega^2\tau_B^2}{(1+\omega^2\tau_B^2)^2}. \quad (28)$$

This function is plotted in Fig. 1. The expression (28) provides the correct value of $\Delta\eta$ at $\omega=0$ [see (3)], whereas at $\omega_0\tau_B=1$ the function $\Delta\eta$ changes its sign, passing from the domain of positive ($\omega<\omega_0$) to the domain of negative ($\omega>\omega_0$) values. In the latter case, it attains the minimum at $\omega_m\tau_B=\sqrt{3}$; at this point $\Delta\eta(\sqrt{3})=-\Delta\eta(0)/8$. At high frequencies of the alternating field the rotational viscosity tends to zero proportionally to ω^{-2}:

$$\Delta\eta=-\frac{\varphi}{\eta}\left(\frac{M_sH}{6\omega}\right)^2 \quad \text{at } \omega\tau_B\gg 1. \quad (29)$$

Here M_s is the magnetization of the ferromagnet; we have used the expression $m=M_sV$ for the magnetic moment of the particle.

B. Second variant

One may arrive at Eq. (28) in a shorter way, performing the computations directly for the linearly polarized field (14). According to Eqs. (7) and (8), the magnetic field, polarized along the x axis, generates in the quiescent fluid (i.e., at $\boldsymbol{\Omega}=0$) the only magnetization component

$$M_x=\chi H_0\cos\alpha\cos(\omega t-\alpha),\quad \tan\alpha=\omega\tau_B. \quad (30)$$

Therefore the magnetic torque $\mathbf{M}\times\mathbf{H}$ vanishes along with the macroscopic angular velocity of the particles $\boldsymbol{\Theta}$ [see (11)]. It was already noted above that the vortical flow destroys the symmetry of the problem, and at $\boldsymbol{\Omega}=(0,0,\Omega)$ the y component of magnetization appears, as well as the z component of the angular velocity of the rotation of the particles. Substitution of $M_y=a\cos(\omega t-\theta)$ into Eqs. (7) and (8) leads to the coupled equations for the amplitude and phase of this component of magnetization:

$$a(\sin\theta-\omega\tau_B\cos\theta)=\Omega\tau_B\chi H_0\sin\alpha\cos\alpha,$$
$$a(\cos\theta+\omega\tau_B\sin\theta)=\Omega\tau_B\chi H_0\cos^2\alpha. \quad (31)$$

Solving this system and taking into account the expression $\tan\alpha=\omega\tau_B$, we find

$$\theta=2\alpha,\quad a=\Omega\tau_B\chi H_0\cos^2\alpha, \quad (32)$$

16 March 2026 08:59:36

so that M_y coincides with the expression (23), derived above. Thus both variants of the solution procedure lead to the same result for $\Delta\eta(\xi,\omega\tau_B)$.

To conclude this section we present a formula for the angular velocity of the particles. Averaging the expression (11) over the period of the field oscillations (that is, over the fast time scale) and employing (26) we obtain

$$\Theta=\Omega\left(1-\frac{\xi^2}{6}\frac{(1-\xi^2\tau_B^2)}{(1+\omega^2\tau_B^2)^2}\right). \tag{33}$$

This formula provides a remarkable illustration of the influence of magnetic field on the viscosity. The permanent or slowly varying ($\omega\tau_B<1$) field hampers the rotation of the particles in the vortical flow. In this case, $\Theta<\Omega$, as can be seen from (33), that is the particles rotate slower than the surrounding fluid. The latter is forced to flow past the particles, which leads to the additional dissipation of the kinetic energy of the fluid. This manifests itself in the additional viscosity $\Delta\eta>0$. With the help of (33), expression (28) may be transformed into

$$\Delta\eta=\frac{3}{2}\,\eta\varphi\,\frac{\Omega-\Theta}{\Omega}. \tag{34}$$

In the absence of the magnetic field the particles rotate together with the fluid ($\Theta=\Omega$), so that no additional friction occurs: $\Delta\eta=0$. On the contrary, the strong stationary field fixes the orientation of magnetic moments of the particles, as a result of which the particles cease to rotate in the flow ($\Theta=0$), and the rotational viscosity reaches the saturation value $\Delta\eta=3\eta\varphi/2$ [see (4)]. In the alternating field for $\omega\tau_B>1$ the particles rotate faster than the fluid, in accordance with (33), that is $\Theta>\Omega$. Now they do not slow down the flow, but, on the contrary, accelerate it: $\Delta\eta<0$! This happens, naturally, because of the transformation of the part of the energy of the alternating magnetic field into the kinetic energy of the fluid.

III. VISCOSITY IN THE ALTERNATING FIELD OF THE ARBITRARY AMPLITUDE

Now we have to give up the linear law of magnetization (15) and solve the nonlinear equation (13):

$$\frac{d\mathbf{M}}{dt}=\boldsymbol{\Omega}\times\mathbf{M}-\frac{1}{\tau_B}\left(\mathbf{M}-M_0\frac{\mathbf{H}}{H}\right)-\frac{\tau_s}{I}\,\mathbf{M}\times(\mathbf{M}\times\mathbf{H}), \tag{35}$$

in which, according to (10), $M_0=nmL[\xi(t)]$. Further on we shall write the time-dependent argument of the Langevin's function in the form $\xi(t)=\xi\gamma(t)$, where ξ is the time-independent dimensionless strength of the field and $\gamma(t)$ is the periodic function with an amplitude equal to unity. In the field $\mathbf{H}=[H(t),0,0]$ under the flow vorticity $\boldsymbol{\Omega}=(0,0,\Omega)$ the magnetization has two components: $M_x=nmF(t)$ and $M_y=nm\Omega\tau_BG(t)$. The ratio of components M_y/M_x is of the order of magnitude $\Omega\tau_B$, since it is obvious that in the quiescent fluid only the x component of magnetization should persist. Now $\Omega\tau_B$ is the only small parameter of the problem. With the first-order accuracy we obtain from (35) the following equations for $F(t)$ and $G(t)$:

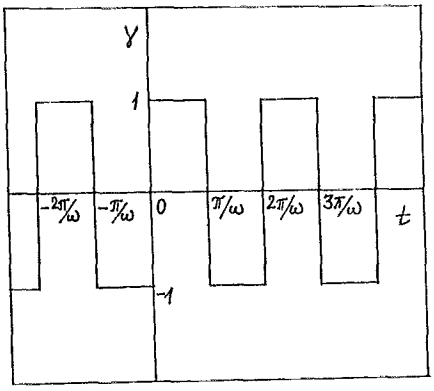


FIG. 2. The graph of the function $\gamma(t)$.

$$\tau_B\frac{dF}{dt}=L[\xi\gamma(t)]-F(t),$$
$$\tau_B\frac{dG}{dt}=F(t)-\left(1+\frac{1}{2}\,\xi\gamma(t)F(t)\right)G(t). \tag{36}$$

They can be solved relatively easily, if the magnetic field is not varying harmonically (14) but in a stepwise way, i.e., when $\gamma(t)$ is a rectangular periodic function with the period $T_0=2\pi/\omega$, as shown in Fig. 2.

We are interested in the value of the magnetic torque $\mathbf{M}\times\mathbf{H}$, averaged over the period of the field. For its only component we have

$$\overline{(\mathbf{M}\times\mathbf{H})_z}=-\overline{M_yH_x}=-3\,\eta\varphi\Omega\,\xi\overline{G(t)\gamma(t)}. \tag{37}$$

Using Eq. (26) one can express the rotational viscosity in terms of this torque with the help of

$$\Delta\eta=\tfrac{3}{4}\eta\varphi\xi\overline{G(t)\gamma(t)}, \tag{38}$$

where

$$\overline{G(t)\gamma(t)}=\frac{\omega}{2\pi}\int_0^{2\pi/\omega}G(t)\gamma(t)dt. \tag{39}$$

Note that according to Eq. (11),

$$\overline{\mathbf{M}\times\mathbf{H}}=-6\,\eta\varphi(\boldsymbol{\Omega}-\boldsymbol{\Theta}).$$

On comparing this equation with (37), we find

$$(\xi/2)\overline{G(t)\gamma(t)}=(\Omega-\Theta)/\Omega.$$

Substitution of the last expression into (38) results in the formula (34). Thus, this formula which was obtained for weak magnetic field is generally true for arbitrary field strengths and frequencies.

In the weak fields ($\xi\ll1$) one may neglect the nonlinear term in the second equation (36), which appears in the equation with the coefficient ξ. The problem is then solved analytically:

$$\Delta\eta=\frac{1}{4}\,\eta\varphi\xi^2\left(2-\tanh\epsilon-2\,\frac{\tanh\epsilon}{\epsilon}\right), \tag{40}$$

where $\epsilon=\pi/2\omega\tau_B$. In the limit $\epsilon\to\infty$ (the stationary field) from (40) it follows, as predicted, $\Delta\eta=\eta\varphi\xi^2/4$. The plot of

16 March 2026 08:59:36

the function $\Delta\eta(\omega\tau_B)$, obtained with the help of expression (40), is qualitatively similar to that given in Fig. 1. The functions (40) and (28) are also quantitatively similar. Actually, under the harmonic variation of the field the rotational viscosity changes its sign at $\omega_0\tau_B=1$ and has a minimum at $\omega_m\tau_B=\sqrt{3}\approx 1.732$, and at the minimum point it has the value $-0.125\eta(0)$. In the case of the rectangular modulation we obtain from (40) $\omega_0\tau_B=0.98$, $\omega_m\tau_B=1.707$, and $\Delta\eta_m=-0.105\eta(0)$.

For the finite-amplitude fields the problem is reduced to the computation of integrals. From Eqs. (36) it follows that $G(t)=-G(t+\pi/\omega)$. Therefore, instead of (39) one may use the expression

$$\overline{G(t)\gamma(t)}=\frac{\omega}{\pi}\int_0^{\pi/\omega}G(t)dt, \tag{41}$$

that is, it is sufficient to know the y component of magnetization only for the half of the field period. The formal solution of Eqs. (36) is

$$G(t)=\frac{e^{g(t)}}{\tau_B}\left(-\frac{e^{g(\pi/\omega)}}{1+e^{g(\pi/\omega)}}\int_0^{\pi/\omega}F(t)e^{-g(t)}dt + \int_0^t F(t')e^{-g(t')}dt'\right), \tag{42}$$

where

$$F(t)=L(\xi)\left(1-\frac{2e^{-t/\tau_B}}{1+e^{-\pi/\omega\tau_B}}\right),$$
$$g(t)=-\frac{t}{\tau_B}\left(1+\frac{1}{2}\xi L(\xi)\right)+\xi L(\xi)\frac{1-e^{-t/\tau_B}}{1+e^{-\pi/\omega\tau_B}}. \tag{43}$$

Integrating the function (42) over the half-period and substituting the result of the averaging (41) into the expression (38), we find the rotational viscosity. The results of the computations are displayed in Figs. 3–5.

Figure 3 presents the dependence of the relative viscosity

$$\Delta\eta(\xi,\omega\tau_B)/\Delta\eta(\infty,0)$$

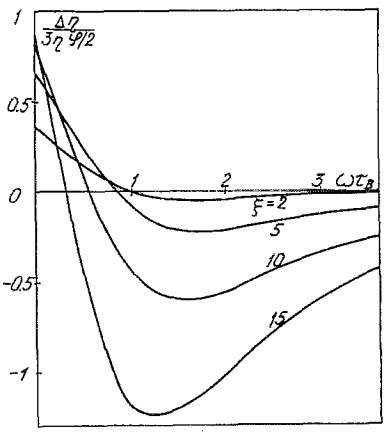


FIG. 3. The relative rotational viscosity as a function of the frequency of the field for several values of its dimensionless amplitude.

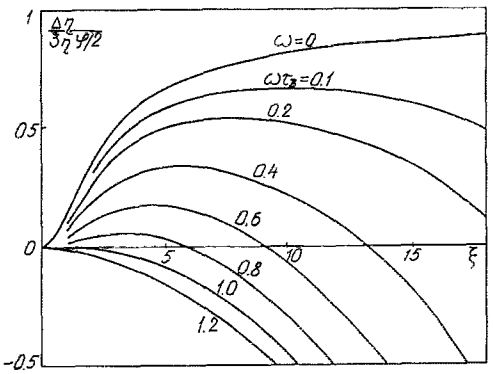


FIG. 4. The relative rotational viscosity as a function of ξ for several values of $\omega\tau_B$.

on the dimensionless frequency of the field $\omega\tau_B$ for several values of its dimensionless amplitude ξ. Qualitatively reproducing Fig. 1, the plots of Fig. 3 show that the frequency ω_0, at which $\Delta\eta$ changes its sign, is decreasing with the growth of ξ: the stronger the field, steeper is the decrease of $\Delta\eta(\omega\tau_B)$. For this reason the dependency $\Delta\eta(\xi)$ proves to be nonmonotonic—see Fig. 4. In the frequency range $0<\omega\tau_B<1$, all the curves in this figure reach the maximal values, before they pass into the domain of negative $\Delta\eta$. It seems curious that in strong magnetic fields the rotational viscosity is negative for arbitrary frequencies. Only in the stationary field the viscosity is always positive and tends to the saturation value (4) at $\xi\to\infty$. This can be seen in the uppermost curve in Fig. 4, which corresponds to $\omega\tau_B=0$ and is given by the formula (2).

Figure 5 presents the diagram of viscosity, where the isolines $\Delta\eta=$const are plotted across the plane $(\xi,\omega\tau_B)$. The solid line splits the plane into two parts: the region of positive values of $\Delta\eta$, lying in the left part, and the right region of negative values. The numbers near the isolines denote the ratio of $\Delta\eta(\xi,\omega\tau_B)$ to the saturation viscosity in the permanent field: $\Delta\eta(\infty,0)=3\eta\varphi/2$. Thus, for instance, the uppermost isoline, denoted with -1, corresponds to the value of $\Delta\eta=-3\eta\varphi/2$.

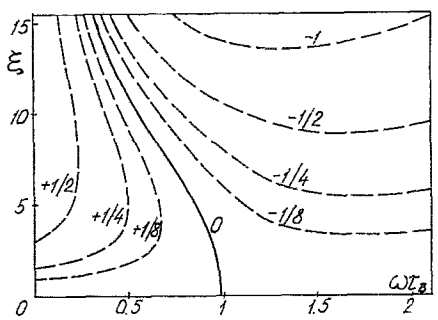


FIG. 5. The map of viscosity $\Delta\eta(\xi,\omega\tau_B)$. Curves present isolines $\Delta\eta=$const. Numbers near the curves mean values $\Delta\eta/(\frac{3}{2}\eta\varphi)$. The solid line corresponds to $\Delta\eta=0$.

16 March 2026 08:59:36

IV. CONCLUSION

The question about the attainable negative values of $\Delta\eta$ is of substantial interest. The answer cannot be deduced from general considerations, since there are no real limits to the magnitude of $\Delta\eta$. In particular, the question about the sign of the total viscosity η_{total} is quite legal. Actually, this is only a question of existence of the isoline $\Delta\eta=-\eta$, or, in the notation of Fig. 5,

$$\Delta\eta(\xi,\omega\tau_B)\Delta\eta(\infty,0)=-2/3\varphi.$$

At the volume concentration $\varphi=0.2$ of the magnetic phase, which is quite realistic, the value $\eta_{\text{total}}=0$ would be matched in Fig. 5 by the isoline $-10/3$. Obviously, it must be located in the domain of high ξ and $\omega\tau_B$. Can these high values really be achieved?

This problem may be approached from another angle by studying the equilibrium stability of the ferrofluid in an alternate magnetic field. Considering the unstable states as states with negative viscosity, we could conclude that the neutral curve of the stability of the equilibrium on the $(\xi,\omega\tau_B)$ plane is exactly the isoline $\eta_{\text{total}}=0$.

In Ref. 5 the stability of mechanical equilibrium was studied for a ferrofluid in a plane channel bounded by immovable rigid walls at $x=0,l$ in the presence of magnetic field $H_x=H_0\cos\omega t$. Both the one- and two-dimensional perturbations of velocity were considered. In the first case, $\mathbf{v}=[0,v(x,t),0]$, which implies $\mathbf{\Omega}=(0,0,\frac{1}{2}\partial v/\partial x)$. The solution of the equations was sought in the form $v=w(t)\sin qx$, $M_y=f(t)\cos qx$ (choosing l the unit of length). The boundary conditions for the velocity $v(0,t)=v(1,t)=0$ determine the spectrum of $q=\pi r$, $r=1,2,3,\ldots$. The most dangerous perturbation corresponds to minimal q, that is $q=\pi$. Such flow provides nonzero flux through the transversal cross section of the channel, so that if ξ and $\omega\tau_B$ are taken from the instability region then the entire system is working as a pump.

For $q=\pi$ and $\varphi=0.2$ the minimum on the neutral curve (i.e., on the critical isoline $\eta_{\text{total}}=0$) corresponds to critical values $\xi_c=126$ and $\omega_c\tau_B$. However, this value of ξ_c requires too high an amplitude of the magnetic field, under which the magnetic particles cannot be treated as rigid dipoles.

ACKNOWLEDGMENTS

We would like to thank Dr. M. Zaks, Dr. T. Felici, and Professor P. N. Kaloni for their attempts to correct our English. One of us (M.I.S.) acknowledges the hospitality in the Pierre and Marie Curie University of Paris. This work was partially supported by the Russian Fundamental Research Foundation under Grant. No. 93-013-17682.

[1] R. E. Rosensweig, *Ferrohydrodynamics* (Cambridge University Press, New York, 1985).

[2] M. I. Shliomis, "Effective viscosity of magnetic suspensions," Sov. Phys. JETP **34**, 1291 (1972).

[3] J. P. McTague, "Magnetoviscosity of magnetic colloids," J. Chem. Phys. **51**, 133 (1969).

[4] M. I. Shliomis, "Magnetic fluids," Sov. Phys. Usp. **112**, 153 (1974).

[5] M. I. Shliomis, T. P. Lyubimova, and D. V. Lyubimov, "Ferrohydrodynamics: An essay on the progress of ideas," Chem. Eng. Comm. **67**, 275 (1988).

16 March 2026 08:59:36